

Historical development and types of memristors

The journey of memristors from a theoretical concept to practical devices involved significant milestones:

- **HP Labs' Breakthrough (2008).** In 2008, researchers at Hewlett-Packard demonstrated the first practical memristor using a thin film of titanium dioxide, confirming Chua's theoretical predictions and paving the way for further research and development.

Since then, various types of memristors have emerged.

- **Resistive RAM (ReRAM).** Utilises memristive properties for non-volatile memory applications
- **Spintronic memristors.** Employ magnetic properties to enhance data storage and processing capabilities
- **Polymeric memristors.** Use organic materials to create flexible and biocompatible devices

again, thanks to the movement of ions or vacancies within the memristor's material, typically a thin film of metal oxide.

Key characteristics of memristors include their non-volatility, allowing them to retain their resistance state even when the power is off, which is essential for memory storage. They are also highly scalable, with the ability to be manufactured at nanoscale dimensions, facilitating high-density integration. Moreover, memristors consume less power compared to traditional memory devices due to their efficient operation.

Neuromorphic computing: An overview

Neuromorphic computing aims to replicate the neural architecture and functioning of the human brain, striving to create more efficient and adaptive computing systems. Traditional von Neumann architectures, which separate processing and memory units, face limitations in speed and power consumption due to the constant shuttling of data between the CPU and memory—a problem known as the von Neumann bottleneck. Neuromorphic systems, however, integrate memory and processing, much like the brain, enabling parallel and distributed computation.

Core components of neuromorphic systems

1. Neurons. Analogous to their biological counterparts, these units process information and perform computations

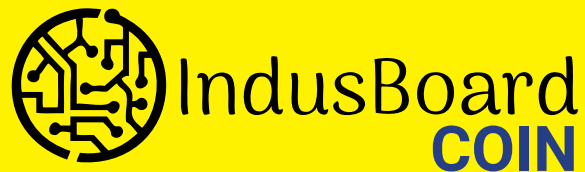
2. Synapses. Connections between neurons, facilitating communication and learning by adjusting their strength, known as synaptic weight



An "AWESOME" Development Board from India?

- **Super Small:** 3 centimeters
- **Super Smart:** 3 processors
(Xtensa® Microprocessor + 2 ULP RISC-V Coprocessor)
- **Super Sensitive:** 4 Sensors
- **Super Connected:** Wifi & 10+ Interfaces
- **Super Flexible:** 30 GPIOs

Introducing the



Priced Shockingly at ₹1999 + GST i.e. ₹2,358



SPECIAL INTRO PRICE

₹1499 + GST i.e. ₹1,768



KNOW MORE / BUY NOW: indus.electronicsforu.com

Developed by the Electronics For You team, for you.